# L-9

18 April, 2025, Friday, 9:30pm

Molecular Clock Hypothesis: Considers that evolutionary rate of two species r same which is not practical.

UPGMA assumes it, which is not practical.

## Neighbour Joining Method:

**Step 1:** $r(i)$ = sum of evolutionary distance from $j=1$ to $n$

$$r(i) = \sum_{j=1}^{n} d(j,i)$$

Ex: For 4 species.

$$r(A) = d(A,B) + d(A,C) + d(A,D)$$
$$r(B) = d(B,A) + d(B,C) + d(B,D).$$

**Step 2:** M table computation.

$$M(i,j) = d(i,j) - \frac{r(i)+r(j)}{n-2}$$

$n$ = no. of species / clusters available at that time

**Step 3:** $$b(u,i) = \frac{d(i,j)}{2} + \frac{r(i)-r(j)}{2n-4}$$ (smallest $(i,j)$ in M) $u=(i,j)$

$$b(u,j) = d(i,j) - b(u,i).$$

**Step 4:** $$d(u,k) = \frac{d(i,k)+d(j,k)-d(i,j)}{2}$$

Input : Evolutional distance matrix

Example:

| | A | B | C | D |
|---|---|---|---|---|
| A | 0 | 3 | 4 | 5 |
| B | 3 | 0 | 5 | 4 |
| C | 4 | 5 | 0 | 7 |
| D | 5 | 4 | 7 | 0 |

Iteration 1:

Step 1:

$r(A) = d(A,B) + d(A,C) + d(A,D) = 3+4+5 = 12$

$r(B) = d(B,A) + d(B,C) + d(B,D) = 3+5+4 = 12$

$r(C) = 4+5+7 = 16$

$r(D) = 5+4+7 = 16$

Step 2:

M's table $\left[ M(i,j) = d(i,j) - \frac{r(i)+r(j)}{n-2} \right]$

| | A | B | C | D |
|---|---|---|---|---|
| A | – | −9 | −10 | −9 |
| B | | – | −9 | −10 |
| C | | | – | −9 |
| D | | | | – |

Step 3: smallest value in M's table is −10. So, there can be cluster of $\underbrace{(A,C) \text{ or } (B,D)}$

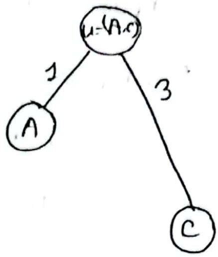
Unlike UPGMA Only 1 cluster will be consider

**Step 3:** Consider (A,C) cluster:

$$b((A,C),A) = \frac{d(A,C)}{2} + \frac{r(A)-r(C)}{2n-4}$$

$$= \frac{4}{2} + \frac{12-16}{2\times4-4}$$

$$= 1$$




$$b((A,C),C) = d(A,C) - b((A,C),A)$$

$$= 4-1$$

$$= 3$$

**Step 4:** Update EDM.

$$d(u,k) = \frac{d(i,k)+d(j,k)-d(i,j)}{2}$$

| | (A,C) | B | D |
|---|---|---|---|
| (A,C) | 0 | 2 | 4 |
| B | 2 | 0 | 4 |
| D | 4 | 4 | 0 |

~~no need~~

$$d((A,C),B) = \frac{d(A,B)+d(C,B)-d(A,C)}{2}$$

$$= \frac{3+5-4}{2} = 2$$

$$d((A,C),D) = \frac{5+7-4}{2} = 4.$$

Iteration 2:

Step 1:

$\pi(AC) = 6$

$\pi(B) = 6$

$\pi(D) = 8$

Step 2:

M's table

| | AC | B | D |
|---|---|---|---|
| AC | – | –40 | –30 |
| B | | – | –30 |
| D | | | – |

here, $n = 3$.

clustering 2 un-clustered species is standard.

Step 3:

consider (B, D) cluster

$$b((B,D),B) = \frac{d(B,D)}{2} + \frac{\pi(B) - \pi(D)}{2n-4}$$

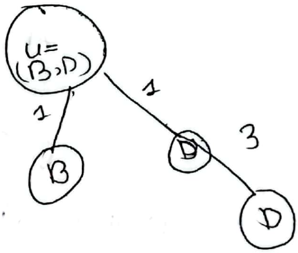
$$= \frac{4}{2} + \frac{6-8}{2\times 3 - 4}$$

$$= 1$$

$$b((B,D),D) = \frac{d(B,D)}{2} = d(B,D) - b((B,D),B)$$

$$= 4 - 1$$

$$= 3$$



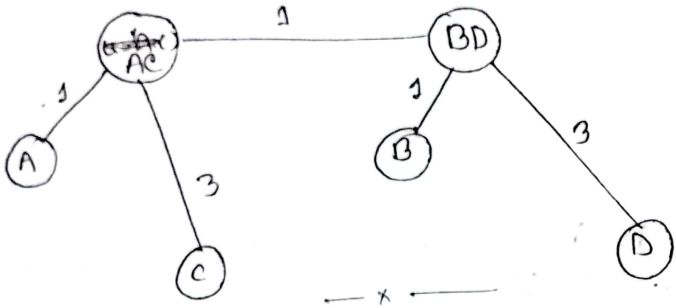

Step 3: considering ( ) cluster

b

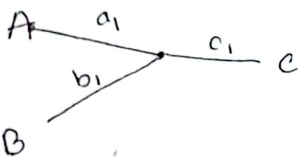
Step 4:

| | AC | (B,D) |
|---|---|---|
| AC | 0 | 1 |
| (B,D) | 1 | 0 |

There is no single common ancestor. It has no roots. Neighbour joining methods produce unrooted Phylogentic tree




— x —




$d(A,B) = a_1 + b_1$ ——(i)

$d(A,C) = a_1 + c_1$ ——(ii)

$d(B,C) = b_1 + c_1$ ——(iii)

(i)+(ii)−(iii)

$a_1 + \cancel{b_1} + a_1 + \cancel{c_1} - \cancel{b_1} - \cancel{c_1} = d(A,B) + d(A,C) - d(B,C)$

3 species formula or 3 point.

$\Rightarrow a_1 = \frac{1}{2}[d(A,B) + d(A,C) - d(B,C)]$

$b_1 = \frac{1}{2}[d(A,B) + d(B,C) - d(A,C)]$

$c_1 = \frac{1}{2}[d(A,C) + d(B,C) - d(A,B)]$

$N_x$ : no. of species in cluster X

~~d(x~~ $d(X,Y) = \frac{1}{N_x N_y} \sum_{i \in x, j \in y} d(i,j)$

L-11 Sun, 27April, 2025, 3:45 pm

Query Sequence (Q): gene $seq^n$ of new species.
we need to match Q with established DB.

Ex: Target Sequence, $T_1$ = A C G T A — length equal হতে হবে
(index 0 1 2 3 4 above A C G T A)

Query Sequence, $Q_1$ = A C T T A

We need

1. Scoring System (findout which 1 target is best for matching)

2. Gap (Insert Nitrogeneous base or delete it)

$T_2$ = A C G T A T — length same না

Q = A C T T A —

A C — T T A

- gap can be added to any position of the $seq^n$
- make the length of two sequence and equal & compare them : Sequence Alignment Problem
- Q can be aligned with multiple T.
- best T which has max score.

L-11 Sun, 27April, 2025, 3:45 pm

Query Sequence (Q): gene seq$^n$ of new species.

we need to match Q with established DB.

Ex: Target Sequence, $T_1$ = A C G T A (0 1 2 3 4) — length equal হলো হল

Query Sequence, $Q_1$ = A C T T A

We need

1. Scoring System (findout which 1 target is best for matching)

2. Gap (Insert Nitrogeneous base or delete it)

$T_2$ = A C G T A T

Q = A C T T A —

A C — T T A — length same না

- gap can be added to any position of the seq$^n$

- make the length of two sequence and equal &

compare them : Sequence Alignment Problem

- Q can be aligned with multiple T.

- best T which has max score.

$\boxed{4}$

| | 0 | 1 | 2 | 3 | |
|---|---|---|---|---|---|
| $Q_1$ - | A | C | T | | Score: |
| $T_1$ - | A | C | T | | ; $Q_1, T_1$ - $1+1+1 = 3$ |
| $T_2$ = | A | G | T | | ; $Q_1, T_2 = 1+0+1 = 2$ |
| $T_3$ - | A | G | T | G | |

— Gap is not a biological event

— Gap penalty: cost to introduce gap.

— length of $T_3$ & Q r not same. to align $T_3$ & $Q_3$ we need to add gap (insert N-base into $Q_1$ or ~~del~~ dlt from $T_3$).
If add gap, we need to consider gap penalty

Match: 1
Mismatch: 0
Gap Penalty: -1

| | 0 | 1 | 2 |
|---|---|---|---|
| $S_1$ = | A | C | T |
| $S_2$ = | A | G | T |

Identity or Identical N-base

↓
Mutation

## Protein:

$S_1$ = Met – Pro – Ala

$S_2$ = Leu – Pro – Ala

Mutation

Conservation (No-change)

here, char. of Met n Leu is same. so, there will be no chg in species' char. genetic dissimilarities কম

যেমন
Met → Leu : probability high & score high
Met → Arg : " & " low as char. of Met r diff.

যেমন
→ PAM score
→ BLOSUM

## Gap Penalty:

① Constant G.P : −2, সবসময় gap থাকুক না কেন penalty same

② Linear G.P : len × ~~G.P~~

③ Affine G.P

Example: A C G – T – G – C

= 3× len × G.P. (GP assume করে নিব)

Example: A C G – – – T G C

Gap opening (−12)

Gap extension [(−1) × len]

A.G.P = −12 + (−1) × 2
= −14

[4]

$S_1$ = A G C T _ T _ T C A

$S_2$ = G G C T A T — T _ A _ _ G

Determine Affine G.P.

Gap opening penalty = −5

Gap extension penalty = −1

A.G.P$(S_1)$ = $-5 + (-5) = -10$

A.G.P$(S_2)$ = $(-5) + (-5) + (-5) + (-1) \times 1$

$= -16$

∴ total A.G.P = $-10 + (-16)$

$= -26$

[4] Question:

S1 : A A G G C T T

S2 : A A G G C

S3 : A A G G C A T

If $S_1 = S_2$ and $S_2 = S_3$, then $S_1 = S_3$ ?

Sequence Identity, SI $(S_1, S_2)$ = 100%

SI$(S_2, S_3)$ = 100%

$$SI(S_1, S_3) = \frac{6}{\min(len(S_1), \min len(S_3))} = \frac{6}{7} \times 100\% = 85.71\%$$

gap. inserted gap, len এ count হবে না,

Assignment: — Markov Model and Hiden Markov Model

L-12 Tues, 29 April, 2025, 2:45 pm

□ S = A C G T

$S_1$ = A G G C

$S_2$ = A G C T

Mutation

Conservation

$S_3$ = T G G C

Given,

$S_1$ : A C G C T A FKI

$S_2$ : G C G C T A FKI

$S_3$ : A G G C T A FKIL

$S_4$ : G C G C T G FKI

G C G C T L F K I

$S_5$ : G C G C T A F K L

$S_6$ : A S G C T A

$S_7$ : A C A C T A F K L

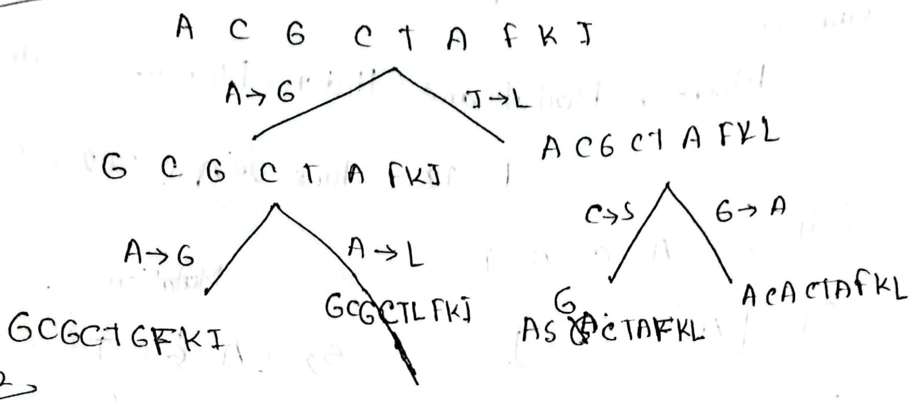
Pairwise alignment

$S_7$:

~~$S_7$~~: Construct PAM 1 Matrix

Step 1:

Step 1:




Step 2

Probability of A get replaced by G.

$P(A \to G)$

$P(x)$ = prob$^{lity}$ of occurance of AA $x$

$$= \frac{\text{no. of } x \text{ occurs in the alignment}}{\text{total no. of amino acid}}$$

$$P(A) = \frac{10}{63}$$

Step 3: $F(x,y) \to$ is the freq. of substitution bet$^n$ $x$ & $y$.

$F(A,G) = F(G,A) = 3$

$F(A,L) = F(L,A) = 1$

$F(C,S) = F(S,C) = 1$

$F(J,L) = F(L,J) = 1$

**Step 4:**

$$f(x) = \sum_{x \neq y} F(x, y)$$

$f(A) = F(A, G) + F(A, L)$

$= 3 + 1 = 4$

$f(C) = F(C, S) = 1$

$f(G) = F(G, A) = 3$

$f(L) = F(A, L) + F(L, J) = 1 + 1 = 2$

$f(J) = F(J, L) = 1$

$f(S) = F(C, S) = 1$

**Step 5:**

$$f = \sum_{x} f(x)$$

$= f(A) + f(C) + f(G) + f(J) + f(L) + f(S)$

$= 4 + 1 + 3 + 2 + 1 + 1$

$= 12$

**Step 6:**

Relativity mutability, $m(x) = \dfrac{f(x)}{100 f \times P(x)}$

$m(A) = \dfrac{4}{100 \times 12 \times P(A)} = \dfrac{4}{100 \times 12 \times 10/63} = 0.021$

A, [illegible] AA দ্বারা replaced by other → m(A).

$$M(A, A) = 1 - m(A)$$
$$= 1 - 0.021$$
$$= 0.979$$

Step 7:

$$M(x, y) = p \cdot \frac{F(x, y)}{f(x)} \times m(x)$$

$$M(A, G) = \frac{F(A, G)}{f(A)} \times m(A)$$
$$= \frac{3}{4} \times 0.021$$
$$= 0.01575$$

Log OR score : $S(x, y) = 10 \log_{10} \frac{M(x, y)}{P(y)}$

score that is x is muted by 'y'.

$$S(A, G) = 10 \log_{10} \frac{M(A, G)}{P(G)} = 10 \log \frac{0.01575}{\frac{10}{63}}$$
$$= -10.0338$$

PAM 1: 1st gen এ 1% AA of seqⁿ will chg into another হওয়ার prob$^{1ty}$

PAM2 = PAM1 × PAM1

PAM10 = $(PAM1)^{10}$

BLOSSUM : to align distantly matched seq$^{n}$

L-13 30April, 2025, Wed, 3:15pm

Tut

Given,

| | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $S_1$ = | B | A | B | A |
| $S_2$ = | A | A | A | C |
| $S_3$ = | A | A | C | C |
| $S_4$ = | A | A | B | A |
| $S_5$ = | A | A | C | C |
| $S_6$ = | A | A | B | C |

Construct BLOSSUM Scoring Matrix.

- to align two different distantly related species (seq$^{n}$) we can use higher value of PAM matric or BLOSSUM matrix

- Conventional BLOSSUM matrix : blossum 45.
- score that is provided to Align A to B. is calculated in BLOSSUM matrix.

## Solution:

### Step 1:

- Count frequency of each AA

| AA | Frequency |
|---|---|
| A | 14 |
| B | 4 |
| C | 6 |
| ~~D~~ | 24 |

### Step 2: Calculate the observe AA pair Substitution:

| | A | B | C |
|---|---|---|---|
| A | 26 | 8 | 10 |
| B | ~~B~~ 8 | 3 | 6 |
| C | 10 | 6 | 7 |

$n(AA) \rightarrow$ A gets replaced by A.

$$n(AA) = \overset{\overset{0}{\downarrow}}{{}^5C_2} + \overset{\overset{1}{\downarrow}}{{}^6C_2} + \overset{\overset{2}{\downarrow}}{0} + {}^2C_2 = 26$$

$$n(AB) = {}^5C_1 \times {}^1C_1 + 0 + {}^1C_1 \times {}^3C_1 + 0 = 8$$

$$n(AC) = 0 + 0 + {}^1C_1 \times {}^2C_1 + {}^2C_1 \times {}^4C_1 = 10$$

$$n(BB) = 0 + 0 + {}^3C_2 + 0 = 3$$

$$n(BC) = 0 + 0 + {}^3C_1 \times {}^2C_1 + 0 = 6$$

$n(CC) = 0 + 0 + {}^{2}C_2 + {}^{4}C_2 = 7$

total no. of pairs possible $= 4 \times {}^{6}C_2 = 60 \rightarrow$ each pair 2 N

4 column → 6 pa seq$^n$

## Step 3:

Calculate observe Prob$^{lity}$ AA pair:

| | A | B | C |
|---|---|---|---|
| A | 26/60 | 8/60 | 10/60 |
| B | 8/60 | 3/60 | 6/60 |
| C | 10/60 | 6/60 | 7/60 |

## Step 4:

Compute Expected frequency.

| | A | B | C |
|---|---|---|---|
| A | 196 | 112 | 168 |
| B | 112 | 16 | 48 |
| C | 168 | 48 | 36 |

| A | A | B | C |
|---|---|---|---|
| A | $\frac{14C_1 \times 14C_1 \times 2!}{2!} = 196$ | $14C_1 \times 4C_1 \times 2! = 112$ | $14C_1 \times 6C_1 \times 2! = 168$ |
| B | ~~196~~ 112 | $\frac{4C_1 \times 4C_1 \times 2!}{2!} = 16$ | $4C_1 \times 2 \times 6C_1 = 48$ |
| C | 168 | 48 | $\frac{6C_1 \times 6C_1 \times 2!}{2!} = 36$ |

Step 5:

Compute Expected Prob$^{lty}$, P(E).

All possible ~~pai~~ expected pair = $24C_1 \times 24C_1$

= 576

| | A | B | C |
|---|---|---|---|
| A | 196/576 | 112/576 | 168/576 |
| B | 112/576 | 16/576 | 48/576 |
| C | 168/576 | 48/576 | 36/576 |

Step 6:

| | A | B | C |
|---|---|---|---|
| A | 0.6975 | -1.0886 | -1.6147 |
| B | | | |
| C | | | |

each value get normalized by $2\log_2\left(\frac{P(O)}{P(E)}\right)$

## Sequence Alignment Problem:

i) Global Alignment Problem: Needleman–Wunch Algo

– compares entire DNA $seq^n$ of two species

ii) Local Alignment Problem

– Smith Waterman Algo

– compares specific gene $seq^n$ of two species

Edit Distance Algo দিয়ে alignment

অংক Given,

$S_1$ = A G C T

$S_2$ = A T G C T

Find Global Alignment using N.W. Algo between $S_1$ & $S_2$

$Sol^n$: Matching $\rightarrow$ 1 len($S_1$) = 4 $\leftarrow$ m

Mismatching $\rightarrow$ -1 len($S_2$) = 5 $\leftarrow$ n

Gap $\longrightarrow$ -2

$\hookrightarrow$ Matrix Size $\therefore$ $(m+1) \times (n+1)$

or, $(n+1) \times (m+1)$

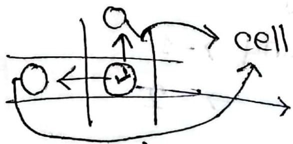
| $S_1$ \ $S_2$ | 0 | 1 A | 2 T | 3 G | 4 C | 6 T |
|---|---|---|---|---|---|---|
| 0 | 0 | -2 | -4 | -6 | -8 | -10 |
| A 1 | -2 | 1 | $-1+(-2)$, $1+(-2)$, $-4+(-2)$ -1 | -3 | -5 | -7 |
| G 2 | -4 | -1 | 0 | 0 | -2 | -4 |
| C 3 | -6 | -3 | -02 | -1 | 1 | -1 |
| T 4 | -8 | -5 | -2 | -3 | -1 | 2 |

[0][1] = " ", "O" → 1টা ~~ope~~ edit operation perform করতে হবে. ∴ [0][1] = 1x gap penalty

$= 1 \times (-2)$

$= -2$

Matrix [1][1] = max $(1+0, -2+(-2), -2+(-2))$



cell এর value সাথে (-2) add

" " " + diagonal

$(-1+-4, -6-2, -1-2)$, $(-1-6, -8-2, -3-2)$

$(-1, -8, -3-2, -10-2)$ $(-1-2, -4-2, 1-2)$

(column) ↗[deletion]

left এ যাওয়া মানে 1st string এ gap add করা

up এ গেলে 2nd string (row) তে gap [insertion]

$S_1$: A — G C T

$S_2$: A T G C T

যদি, PAM matrix / BLOSSUM Matrix দেয়া থাকে তাহলে matching, mismatching value টা PAM/BLOSSUM থেকে নিতে হবে।

Ex:

| | A | T |
|---|---|---|
| A | 0.2 | −1 |
| T | −1 | 4 |

AA matching : 0.2

TT " : 4

AT mismatching : −1

gap penalty will be given.

[4] Local Alignment Problem.

$S_1$: G A C T T A C

$S_2$: C G T G A A T T C A T

Find Local Alignment b/w $S_1$ & $S_2$ using S.W Algo.

Given,

Matching = 5

Mismatching = −3

Gap = −4

max (mismatch, ~~side~~ 0)

| $S_1$ \ $S_2$ | | 0 | 1 C | 2 G | 3 T | 4 G | 5 A | 6 A | 7 T | 8 T | 9 C | 10 A | 11 T |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | G | 0 | 0 | 5 | 1 | 5 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | A | 0 | 0 | 1 | 2 | 1 | 10 | 6 | 2 | 0 | 0 | 5 | 1 |
| 3 | C | 0 | 5 | 1 | 0 | 0 | 6 | 7 | 3 | 0 | 5 | 1 | 2 |
| 4 | T | 0 | 1 | 2 | 6 | 2 | 2 | 3 | 12 | 8 | 4 | 2 | 6 |
| 5 | T | 0 | 0 | 6 | 7 | 3 | 0 | 0 | 8 | 17 | 18 | 9 | 7 |
| 6 | A | 0 | 0 | 0 | 3 | 4 | 8 | 5 | 4 | 13 | 14 | 18 | 14 |
| 7 | C | 0 | 0 | 1 | 0 | 0 | 4 | 5 | 2 | 9 | 18 | 14 | 15 |

max (up ~~value~~ 0, mismatch)

cell value = max( match/mismatch value + diagonal,
left + gap, up + gap, 0 )

- start from max value
- match হলে diagonal
- mismatch হলে go to max (left, up)
  - left এ গেলে $S_1$ এ gap insert, align $S_2$
  - up এ গেলে $S_2$ তে gap, align $S_1$
  - left = up হলে, যেকোনো একটা

$S_1$ = T/A

$S_2$ = T/A

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| $S_1$ = | G | A | C | – | T | T | T | – | A |
| $S_2$ = | G | A | – | A | T | T | T | C | A |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| $S_1$ = | G | A | C | – | T | T | – | A |
| $S_2$ = | G | A | – | A | T | T | C | A |

- যদি অনেকগুলো alignment হয় তাহলে all max value থেকে start.